PROJECT BRIEF

Network Log Translator

Turn raw network logs into plain English — instantly

1. Project Identity

Project Name	Network Log Translator
Tagline	Turn raw network logs into plain English — instantly
Team Name	Binary Bandits
Team Members	Ayush Krishnan P | Hithashree P
Track / Category	Cybersecurity / Developer Tools
Deployed URL	https://h2-h-binary-bandits-network-log-tra.vercel.app/
Repository	github.com/Ayush75-arch/H2H-BinaryBandits-Network_Log_Translator

2. Problem Statement

The Core Problem

Network logs are the ground truth of what is happening inside any system — but they are written for machines, not humans. A single security event scatters context across dozens of raw, unreadable lines:

```
Jun 10 14:23:01 webserver01 sshd[1234]: Failed password for root from 103.45.67.89
port 58321
```

SOC analysts spend minutes to hours manually correlating these entries — figuring out what happened, how severe it is, and what to do. For non-technical stakeholders, the logs are completely inaccessible.

The delay between an event occurring and someone actually understanding it is where breaches slip through.

Key pain points:

- Manual log analysis is slow — a realistic incident can take 15–30 minutes to triage by hand
- Context is fragmented across log types — Syslog, SNMP, and VPC Flow Logs don't speak to each other
- Severity is not obvious — distinguishing a critical attack from routine noise requires expertise
- Non-technical teams cannot participate — incident response becomes a bottleneck on a few senior analysts

3. Proposed Solution

Network Log Translator is an end-to-end pipeline that ingests raw Syslog, SNMP trap, and AWS VPC Flow logs and converts them into structured, human-readable incident reports in seconds.

The system does not just parse logs. It:

- Detects anomalies using both rule-based logic and statistical methods
- Correlates related events from different log lines into unified incidents
- Classifies attack chains using kill-chain methodology
- Scores risk on a 0–100 scale with confidence percentages
- Generates plain-English explanations via LLaMA 3.1, readable by anyone
- Answers follow-up questions about the analysis in natural language

The defining metric we built around: Time-to-Clarity (TTC) — the time from raw log input to an actionable, human-readable insight. Our system reduces this from minutes to seconds.

4. Tech Stack

Layer	Technology	Purpose
Backend	Python 3.11 + FastAPI	API server, pipeline orchestration
AI / LLM	Groq API — LLaMA 3.1 8B Instant	Plain-English log explanations
Log Parsing	Custom regex engine	Syslog, SNMP, VPC Flow extraction
Anomaly Detection	Rule-based + Isolation Forest	Threat flagging and scoring
Frontend	React 18 + Vite + Tailwind CSS	Dashboard, streaming UI
Streaming	Server-Sent Events (SSE)	Live result delivery
Deployment	Vercel (FE) + Render (BE)	Production hosting

5. Core Features

- Multi-format log parsing — Syslog, SNMP traps, and AWS VPC Flow Logs supported out of the box
- Real-time anomaly detection — brute-force attempts, port scans, traffic spikes, repeated auth failures
- Incident correlation — related log entries grouped into named incidents with attack chain classification
- AI-powered explanations — per-entry and per-incident plain-English summaries generated by LLaMA 3.1
- Risk scoring — every log and incident assigned a severity (INFO to CRITICAL) with a 0–100 risk score
- Natural language querying — ask questions about the analyzed logs via a chat-style interface
- Live streaming responses — results appear progressively via SSE, no batch waiting
- Host compromise detection — cross-references failed and successful auth events to surface lateral movement
- Time-to-Clarity benchmarking — built-in tool to measure and display end-to-end pipeline speed

6. Architecture / System Flow

End-to-End Pipeline

```
Raw Log Input → Parser → Anomaly Detector → Classifier → Pipeline → LLM Summarizer → API → React Dashboard
```

Stage-by-stage breakdown:

- Log Parser — custom regex engine extracts timestamp, source IP, host, protocol, and event from raw text across all three supported log formats
- Anomaly Detector — rule-based checks for brute-force thresholds (≥3 failures/min), port scan patterns, traffic volume spikes, plus optional Isolation Forest for statistical outliers
- Classifier — tags every parsed entry with a severity level: INFO, LOW, MEDIUM, HIGH, or CRITICAL
- Pipeline (v7) — cross-event correlation groups related entries by source IP and time window, detects potential compromises, classifies attack chains using kill-chain stages, computes confidence and risk scores (0–100)
- LLM Summarizer — packages structured incident context and calls Groq's LLaMA 3.1 8B Instant to generate human-readable explanations per log and per incident
- FastAPI Backend — exposes /analyze, /stream/logs (SSE), /query, /benchmark, and /context endpoints; persists last analysis context in memory for follow-up queries
- React Frontend — streams results live, renders log table with severity highlighting, incident modals, query panel, and benchmark view

7. Setup Instructions

Prerequisites

- Python 3.11 or higher
- Node.js 18 or higher
- A free Groq API key from console.groq.com

Step 1 — Clone the Repository

```
git clone https://github.com/Ayush75-arch/H2H-BinaryBandits-Network_Log_Translator.git
```

Step 2 — Backend Setup

```
cd backend pip install -r requirements.txt
```

Create a .env file inside the backend/ directory:

```
GROQ_API_KEY=your_groq_api_key_here
```

Start the backend server:

```
uvicorn app:app --reload --port 8000
```

Step 3 — Frontend Setup

```
cd ../frontend npm install npm run dev
```

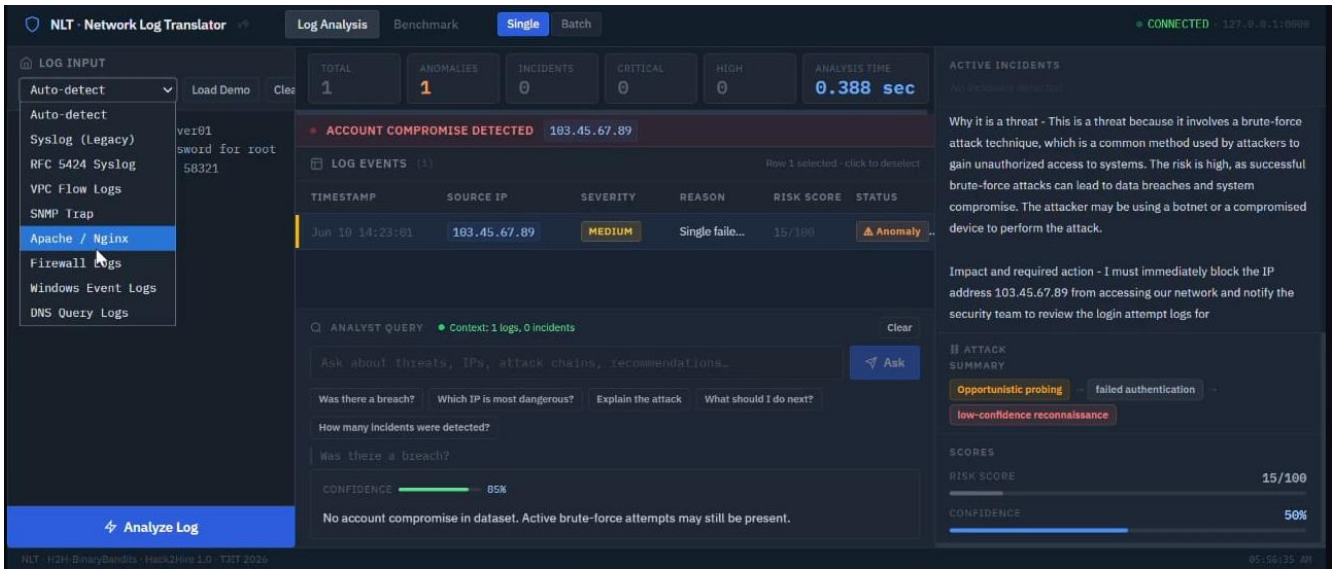
The app runs at http://localhost:5173. The frontend proxies API calls to the local backend on port 8000.

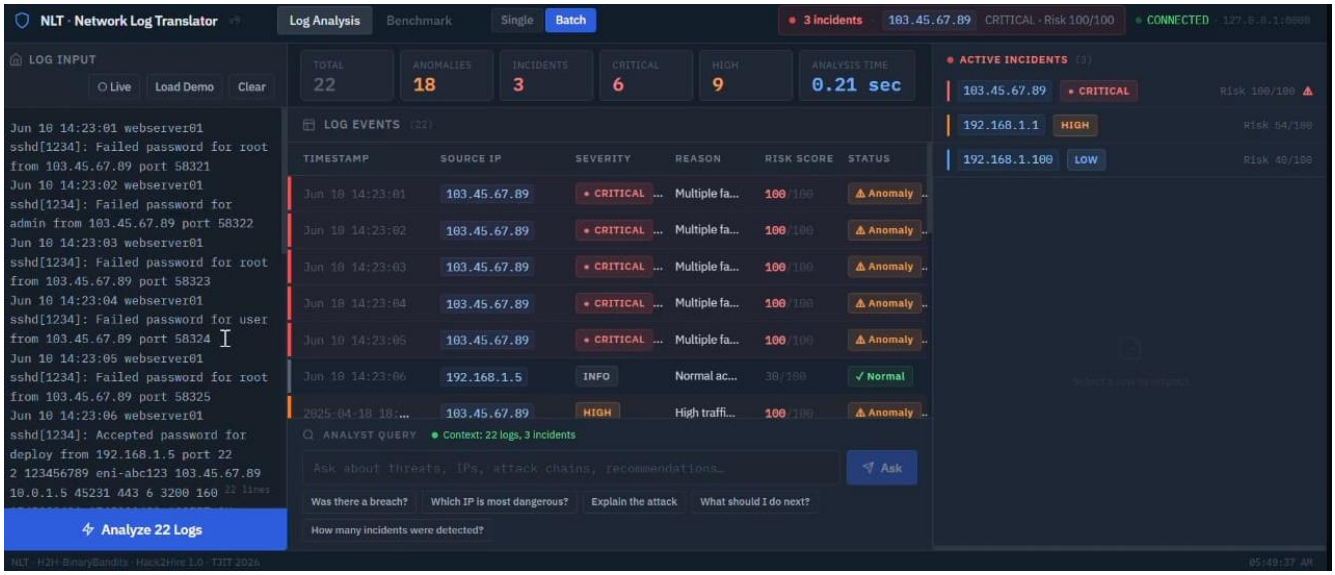
8. Team Members

Name	Role	GitHub
Ayush Krishnan P	Backend, Pipeline, AI Integration	@Ayush75-arch

Hithashree P	Frontend, UI/UX, Integration	@HITHASHREE-GIT

SOME SCREENSHOTS OF THE DEMO:





9. Deployed Links

Frontend (Vercel)	https://h2-h-binary-bandits-network-log-tra.vercel.app/
Backend (Render)	https://h2h-binarybandits-network-log-translator.onrender.com
GitHub Repo	github.com/Ayush75-arch/H2H-BinaryBandits-Network_Log_Translator
YouTube Link	https://youtu.be/bo1eeIC0isc